

S A Q I B A L I

ELECTRICAL ENGINEERING STUDENT | GROUND STATION ENGINEER

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🔗 github.com/WhiteScholar-01

🌐 [Portfolio](#)

PROFILE

Electrical Engineering undergraduate at ZHCET, AMU who takes control designs from simulation to tested hardware. Ground Station Engineer at Orbitalink (a Ground Station as a Service startup), leading a two-axis satellite antenna tracker from RF and chassis to firmware; research intern at IIT Ropar on field-oriented control of a five-phase IPMSM. Chairperson, IEEE Student Branch, AMU. Open to roles in motor drives, control systems, embedded systems, robotics and ground station/RF engineering.

EDUCATION

Zakir Husain College of Engineering & Technology (ZHCET), AMU, Aligarh Sep 2023 – Jun 2027
B.Tech in Electrical Engineering

Syedna Tahir Saifuddin (STS) School, AMU, Aligarh Aug 2020 – Aug 2022
Senior Secondary School Certificate (PCM: Physics, Chemistry, Mathematics) | Grade: A+

PROFESSIONAL EXPERIENCE

Research Intern Jun 2026 – Jul 2026
Indian Institute of Technology (IIT) Ropar, Department of Electrical Engineering Rupnagar, Punjab

- Developed generalized mathematical models of a five-phase interior permanent magnet synchronous motor (IPMSM) and transformed them into rotating reference frames with Clarke–Park transforms in MATLAB/Simulink.
- Designed dual-plane field-oriented control (FOC) in the dq and xy subspaces with PI current and speed loops, loss-aware MTPA (Newton–Raphson solver), flux weakening and five-phase SVPWM; also studied DTC.
- Generated ePWM signals on a TI C2000 DSP in Code Composer Studio, tested the gate driver circuit, and submitted a research report on the work.

Ground Station Engineer (Part-time) Aug 2025 – Present
Orbitalink Private Limited Aligarh, UP

- Lead end-to-end development and integration of a two-axis (azimuth–elevation) antenna tracking system for Orbitalink's Ground Station as a Service platform, from hardware design to firmware.
- Designed a circularly polarised 435 MHz cross-Yagi in MMANA-GAL/NEC with 50 Ω matching, and the tracker chassis in Fusion 360.
- Wrote firmware for NEMA23 hybrid servo motors on HSD57 drives (trapezoidal motion ramps, EEPROM persistence, serial command protocol); porting it from Arduino to STM32F411.
- Managed the field unit's physical infrastructure and cable routing, and resolved hardware–software bottlenecks during system integration.

Electrical Design Intern Jun 2025 – Jul 2025
Orbitalink Private Limited (in collaboration with AMU Innovation Foundation) Aligarh, UP

- Developed and implemented an adaptive PID controller to stabilise and precisely point a high-inertia antenna tracking system.
- Built the dual-axis control loop on an Arduino Uno with AS5600 magnetic encoders over I2C, applying system modelling and performance optimisation for satellite communication infrastructure.

Member, Ground Station Team Sep 2024 – Present
SS AMU SAT (university satellite project) Aligarh, UP

- Joined as Antenna Tracker Volunteer (Sep 2024 – Jul 2025); built the 3D chassis model for the antenna tracker and developed its two-axis PID positioning logic.
- Test tracking mechanisms, support ground station operations and calibrate communication equipment; hands-on with signal reception, hardware troubleshooting and diagnostics.

SKILLS

- Control & Motor Drives:** Field-Oriented Control (FOC), MTPA, flux weakening, DTC, SVPWM, adaptive PID, MRAC, mathematical modelling

- **Power Electronics:** DC-DC converters, inverter PWM, gate driver testing, Simscape Electrical
- **Embedded Systems:** C/C++, STM32 (STM32CubeIDE, CubeMX), TI C2000 (Code Composer Studio), Arduino, 8051, I2C, UART, servo drives
- **RF & Ground Systems:** Yagi and cross-Yagi antenna design, MMANA-GAL, NEC, GNU Radio, SDR, satellite tracking
- **Software & Data:** MATLAB, Simulink, Python (NumPy, Pandas, Matplotlib, scikit-learn), machine learning, OpenCV, JavaScript, Node.js, Git
- **Design Tools:** Fusion 360 (3D CAD, FEA), Altium Designer, KiCad
- **Soft Skills:** Leadership, team management, communication, problem solving, team collaboration

PROJECTS

UHF Satellite Image Transfer & SDR Antenna Testing | GNU Radio, SDR, Python 2026

- Built and validated an end-to-end software-defined radio link at 435 MHz for wireless image transfer, with BPSK modulation pipelines in GNU Radio and custom Python telemetry scripts.

STM32 Test Board | Altium Designer, PCB design 2026

- Designed a compact STM32 board with USB-C power delivery, an onboard buck converter and integrated motion sensing, using hierarchical schematics.

437 MHz 9-Element Long-Yagi (DL6WU) | Antenna design, 70 cm band 2026

- Designed a nine-element long-Yagi that feeds 50 Ω directly through a 1:1 balun, documented as a full construction manual.

Simulink Modelling and Analysis of a DC-DC Boost Converter | MATLAB, Simulink, Simscape 2025

Electrical | File: Model_Boost.slx

- Modelled a boost converter and analysed its steady-state and transient response under varying duty cycles and load conditions (power electronics and SMPS fundamentals).

Weather Dashboard Application | Live Deployment | Source Code (GitHub) | OpenWeatherMap 2024

API, Node.js, Axios

- Built a responsive weather app with real-time temperature, humidity and wind data for any city, a search function and error handling for API failures and invalid input; deployed on Render.

CERTIFICATIONS

- **GE Aerospace – Explore Electrical Engineering Job Simulation**, Forage (Jan 2026)
- **Data Science Course**, CodeWithHarry (Jan 2026)
- **MATLAB Simulation and Control**, IEEE Student Branch, AMU (Oct 2025)
- **Advanced Aircraft Design Workshop**, Bharat Space Education Research Centre (Oct 2025)
- **Certificate of Internship**, Orbitalink Private Limited (Jul 2025)
- **OpenCV Bootcamp**, OpenCV University (Jun 2024)
- **SS AMU SAT Workshop**: Fusion 360 product design and KiCad embedded design for satellites (Jun 2024)
- **Develop GenAI Apps with Gemini and Streamlit; Prompt Design in Vertex AI**, Google (May 2024)
- **Introduction to Cybersecurity**, IEEE Student Branch, AMU

LEADERSHIP & ACTIVITIES

Chairperson, IEEE Student Branch, AMU Aug 2026 – Present

- Lead the student branch and its office-bearer team, managing technical events, workshops and member orientation.

India Mobile Congress

- Represented Orbitalink and showcased the company's field unit at the event.



LinkedIn



GitHub



Portfolio



Email